

EIGHT COMPLEMENTARY SCREENS

Association, paired model difference, and error-local timing answer different questions

REGRESSION

association · global
failures vs retained successes

REGRESSION

association · local
same contrast near first divergence

REGRESSION

paired · global
specialized text – base text

REGRESSION

paired · local
paired difference near divergence

IMPROVEMENT

association · global
improvements vs remaining base failures

IMPROVEMENT

association · local
same contrast near first divergence

IMPROVEMENT

paired · global
specialized text – base text

IMPROVEMENT

paired · local
paired difference near divergence

KNOWN FEATURES RE-APPEAR, BUT IN DIFFERENT CELLS

DSTK100 10168: rank #2 in improvement association (active fraction, E/V -7.67). CodeLlama 27: rank #45 in local paired regression.

New candidate 13147: rank #1 in global regression association (mean, E/V 5.24, support 95).

$\alpha=3$ SWEEP: MANY DIRECTIONS MOVE THE SAME TASKS

A broad causal screen exposed task sensitivity—and the need for placebo directions

DSTK100 · 30 selected features

Baseline 15 / 80
25 / 30 features had positive net change
Best arm: 3048, +4 net
29 / 30 corrected task /1030

CodeLlama · 37 selected features

Baseline 0 / 50
11 / 37 produced one official pass
10 / 11 corrected the same task: /119
13147 alone corrected /490

INTERPRETATION

A pass transition is causal evidence that the residual perturbation mattered—but not yet that the selected latent was specific.

Repeated correction of the same tasks motivated orthogonal shams, matched latent controls, bidirectional steering, and dose curves.

TWO CANDIDATES, TWO SEMANTIC PROFILES

Natural TopK activations and CrossCoder geometry suggest different mechanisms

DSTK100 FEATURE 3048

TOP TOKENS

_ · test · Func · from · Task · import · def

CONTEXTS

helper functions · test blocks · imports
transitions into additional code units

GEOMETRY

decoder cosine base↔FT: 0.961
active in 1,427 / 2,608 texts

CODELLAMA FEATURE 13147

TOP TOKENS

indentation · newline · # · ... · TODO · pass · return

CONTEXTS

Hint: Use ... · placeholders
incomplete or scaffolded implementations

GEOMETRY

decoder cosine base↔merged: 0.275
active in 718 / 2,608 texts

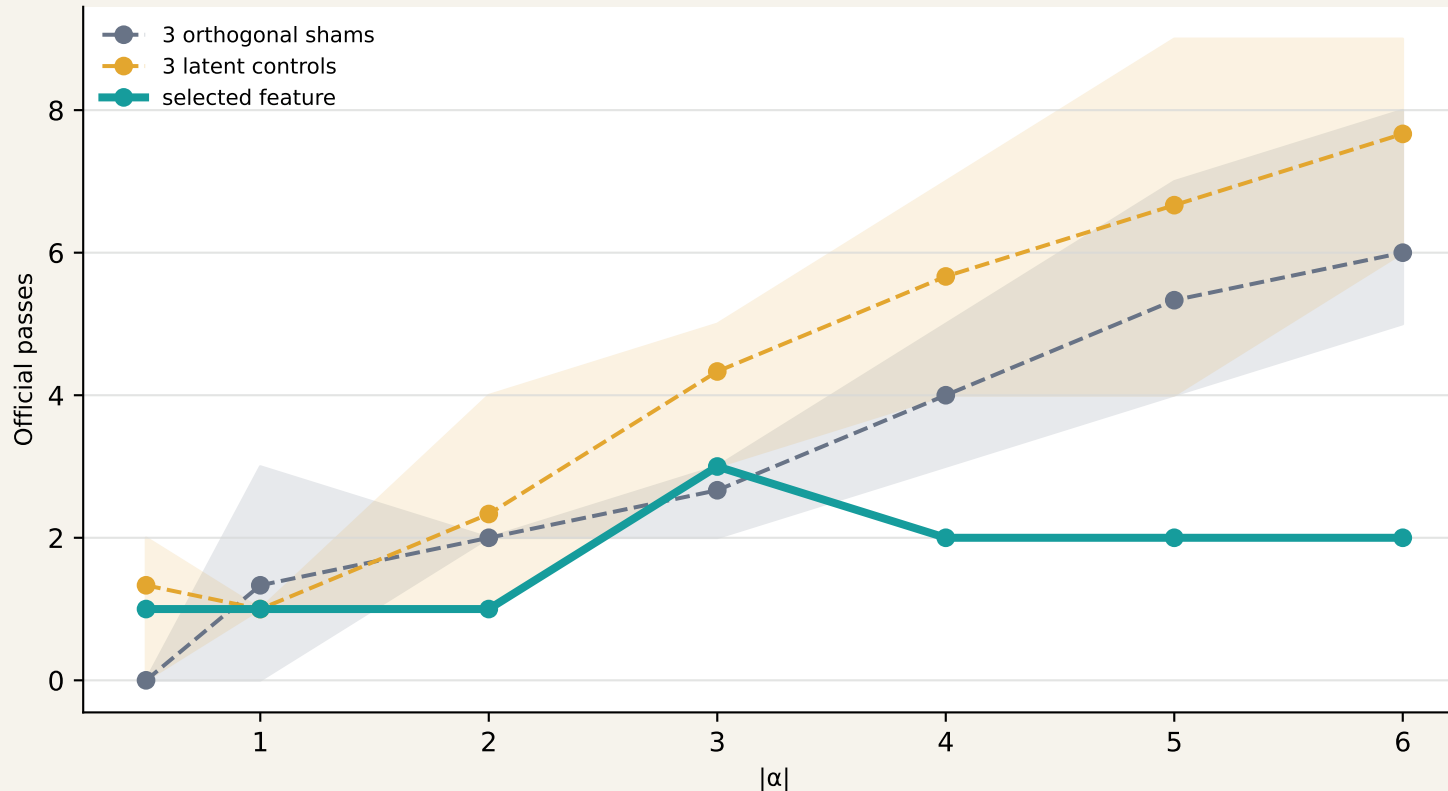
3048 shared program-structure direction; semantically broad

13147 model-differential incomplete-implementation direction

FEATURE 3048: THE EFFECT IS NOT SPECIFIC

Seed 50000 · DeepSeek base · 80 selected contamination improvements

FAIL → PASS IN THE BASE MODEL



At $|\alpha|=3$: 3048=3 passes · shams=2-3 · latent controls=3-5

The selected direction changes behavior, but matched and random perturbations are equally or more effective.

OUTPUTS CHANGED

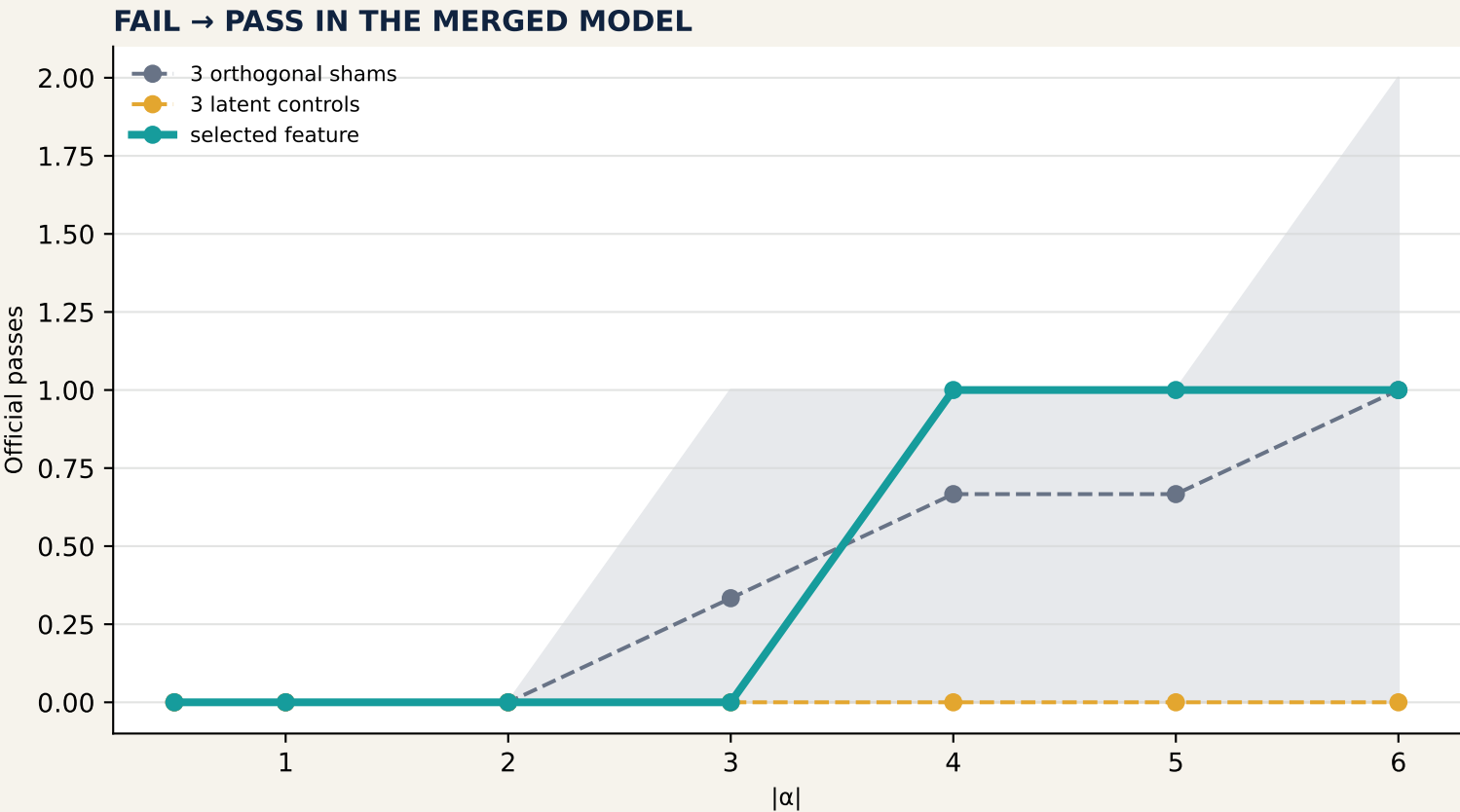
3048: 13 → 37 / 80 across dose
At $|\alpha|=3$: 27 changed
8 test-marker changes
4 import changes
8 other logic/text changes

REVERSE DIRECTION

Adding 3048 to FT:
60 → 55 passes by +5/+6
6 → 40 outputs changed
No reverse sham control yet

FEATURE 13147: COHERENT, BUT PLACEBO-SENSITIVE

Seed 50000 · CodeLlama merged · 50 selected regressions



SEMANTIC CHANGE

13147 changes 8 → 35 / 50 outputs
Return-line changes: 3 → 18
At -4/-5/-6, /490 gains
parse → dump JSON → return result

REVERSE DIRECTION

Adding 13147 to base:
50 → 47 passes at +6
7 → 25 outputs changed
Losses emerge from +2

Matched latent controls: 0 passes at every dose · Shams: up to 2 passes

The semantic match is strong, but one sham also repairs /490 at -6; specificity remains exploratory.

THE 116 ARMS ALTER MORE THAN PASS / FAIL

Exact extraction-v4 evaluated code reveals broad, dose-dependent trajectory changes

3048 FAMILY · 80 TASKS

SELECTED FEATURE, $-0.5 \rightarrow -6$
changed outputs: 13 $\rightarrow$ 37
test-marker changes: 3 $\rightarrow$ 9
import changes: 4 $\rightarrow$ 9
other logic/text: 3 $\rightarrow$ 7

SHAMS AT -6
29-39 outputs changed · 5-8 passes

LATENT CONTROLS AT -6
42-47 changed · 6-9 passes

13147 FAMILY · 50 TASKS

SELECTED FEATURE, $-0.5 \rightarrow -6$
changed outputs: 8 $\rightarrow$ 35
return changes: 3 $\rightarrow$ 18
other logic/text: 2 $\rightarrow$ 11

SHAMS AT -6
26-30 changed · return changes 14-18

LATENT CONTROLS AT -6
19-24 changed · 0 passes

RULE-BASED CHANGE TAXONOMY

test markers $\rightarrow$ imports $\rightarrow$ returns $\rightarrow$ function structure $\rightarrow$ comments $\rightarrow$ other logic/text (mutually exclusive, first matching category)

Steering effects are real and structured, but high-dose output change is not itself evidence of latent specificity.

REVISED EVIDENCE LADDER

Controls separate trajectory control from feature-specific causal explanation

SUPPORTED	EXPLORATORY	NOT SUPPORTED
CrossCoder screens recover stable behavioral contrasts. Selected directions causally alter code trajectories and official outcomes.	13147 has a coherent meaning↔failure-mode link and a stable /490 repair robust performance across 50000 controls.	4904 repairs robust performance across 50000 controls. A pass transition alone does not establish semantic specificity.

CRITICAL REPLICATION NOTE

The discovery $\alpha=-3$ run used seed 1000+task_idx (1490 for /490). The focused sweep used input seed 50000.

Seed 1490 repaired /490 at -3; seed 50000 required -4 to -6. These are robustness runs—not exact replications.

NEXT CONFIRMATORY TEST

Re-run baseline, target feature, 3 shams, and 3 latent controls with the canonical seed convention and multiple paired seeds per α .

Feature 6404 remains the strongest controlled causal result in the project.